Welcome to Machine Learning for Security

Prof. Gilad Gressel
Amrita School of Engineering

Introduction

- About me
- Why machine learning for security
- How I teach
- What I hope for you

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Me

- Gilad Gressel
  - Gill-Odd
- Masters in Computer Science
- Linguistics
- Teaching Forever

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Why Machine Learning for Security

- It's powerful and very relevant today
  - More data than ever
  - More attacks than ever
- Security needs detection
  - Spam
  - Malware
  - Botnets
- Automate detection with Machine learning

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How I Teach

- Motivate the problem
- Think for yourself about a solution
- Fundamentals
- Fundamentals
- Fundamentals

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What I Hope For You

- Enjoy the journey
- Struggle but don't give up ever
- Understand deeply
- Ask a lot of questions!
- But only ask questions after you thought about it

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Conclusion

- I love ML and Security
- We are going to learn fundamentals
- I hope you love the journey
- Ask questions!

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Learning from Patterns

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Amrita School of Engineering

Introduction

- Waiting in line
- Inference Learning
- Learning from data

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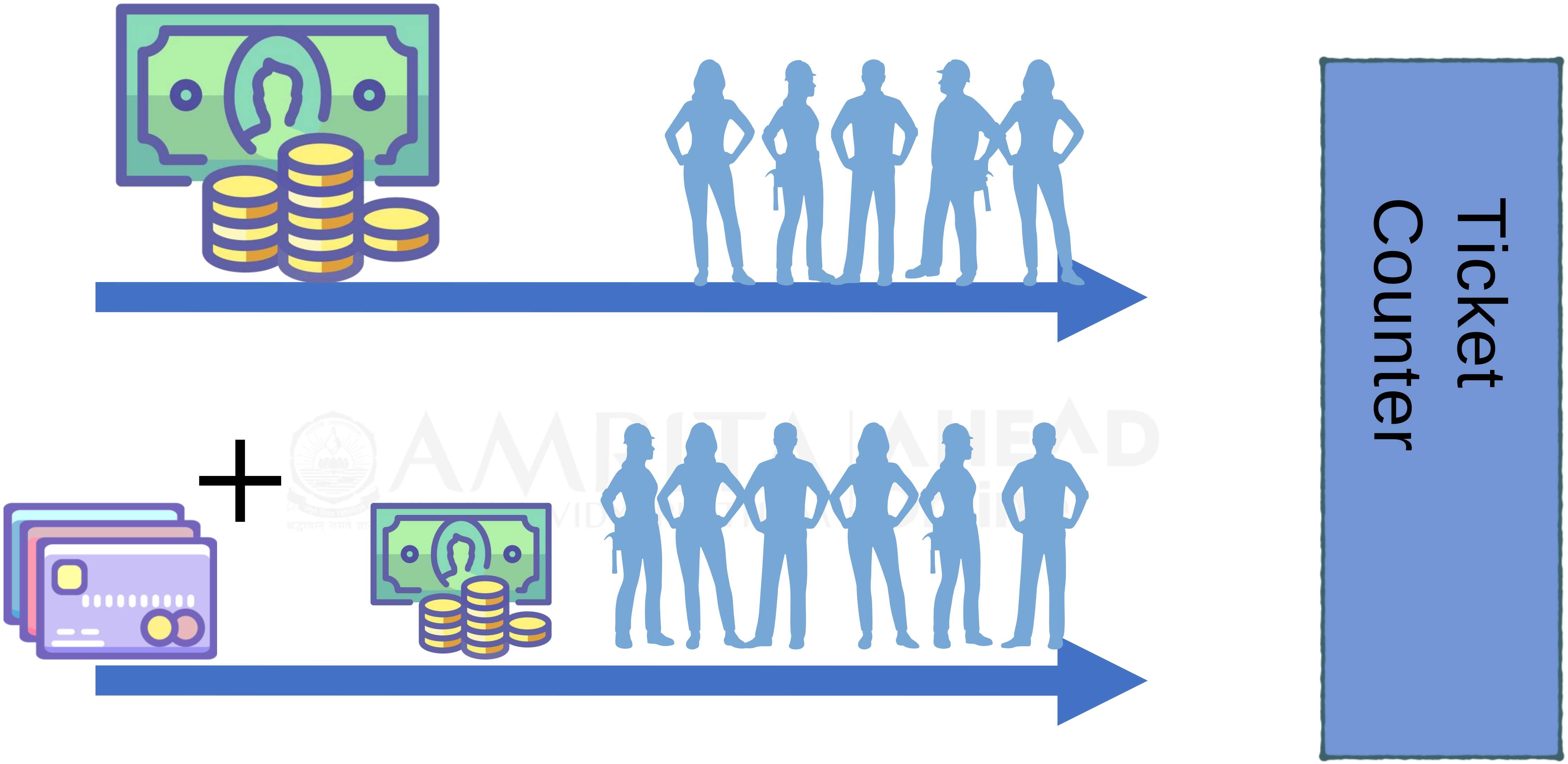
Ticket Counter

Inference learning

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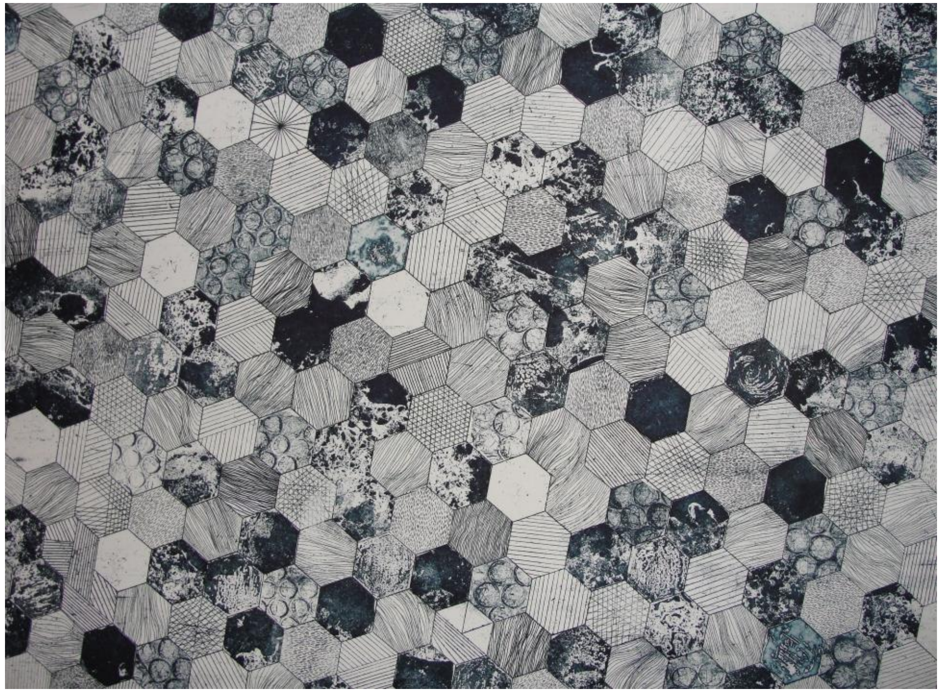
- Given the pattern, learn the rules
- Machine Learning works on the same principles of inference
  - Data provides the pattern
  - Machine Learning learns the rules



Conclusion

- We observed patterns
- We analyzed the patterns and extracted a rule
- This is inference learning
- Machine learning is all about automating this (with machines)

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What is Artificial Intelligence?

Prof. Gilad Gressel
Amrita School of Engineering

Introduction

- What can AI do?
- Examples
- AI vs ML vs DL

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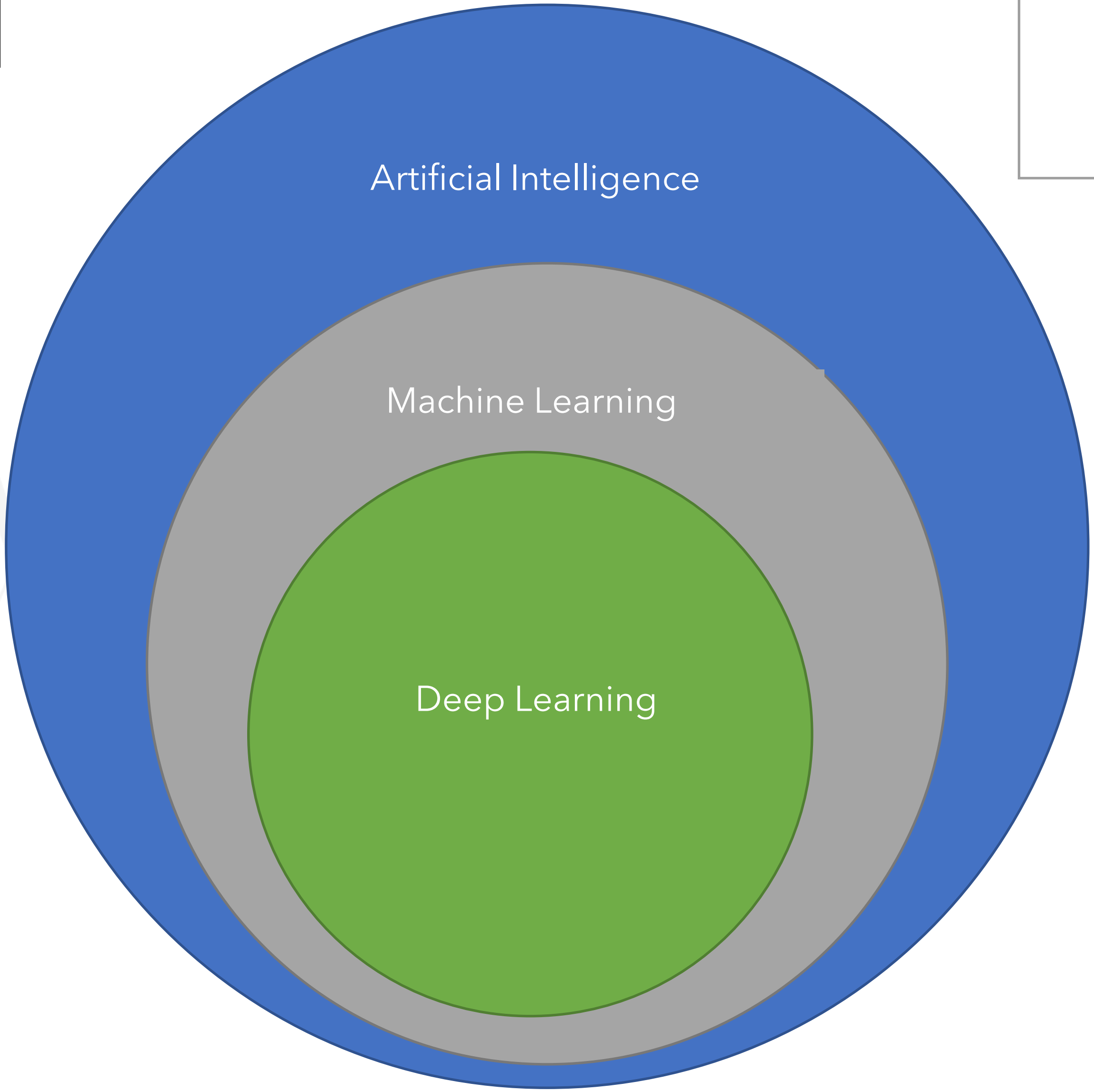
What does Artificial Intelligence Do?

- Spam Detection
- Advertisement placement
- AI Assistants
  - Siri, Google now, Cortana, Alexa
  - Speech to text
- Machine Translations (Google translate)
- Computer vision
  - Facial recognition
  - https://quickdraw.withgoogle.com
- Style Transfer
  - Everybody Dance Now - https://arxiv.org/pdf/1808.07371.pdf

The world of AI

- Machine learning is a subset of AI
- Deep learning is a subset of ML



Conclusion

- Artificial intelligence is everywhere in your life
- It's rapidly evolving
- Machine Learning is a subset of AI
  - It contains within it deep learning

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How do Computers Learn?

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Introduction

- How do computers learn?
- Inductive vs deductive learning
- ML is inductive

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What is 'Learning' ?

- Study
- Experience
- Being taught

learning

/ˈləːnɪŋ/

noun

noun: **learning**

the acquisition of knowledge or skills through study, experience, or being taught.

"these children experienced difficulties in learning"

synonyms: study, studying, education, schooling, tuition, teaching, academic work, instruction, training; More

- knowledge acquired through study, experience, or being taught.

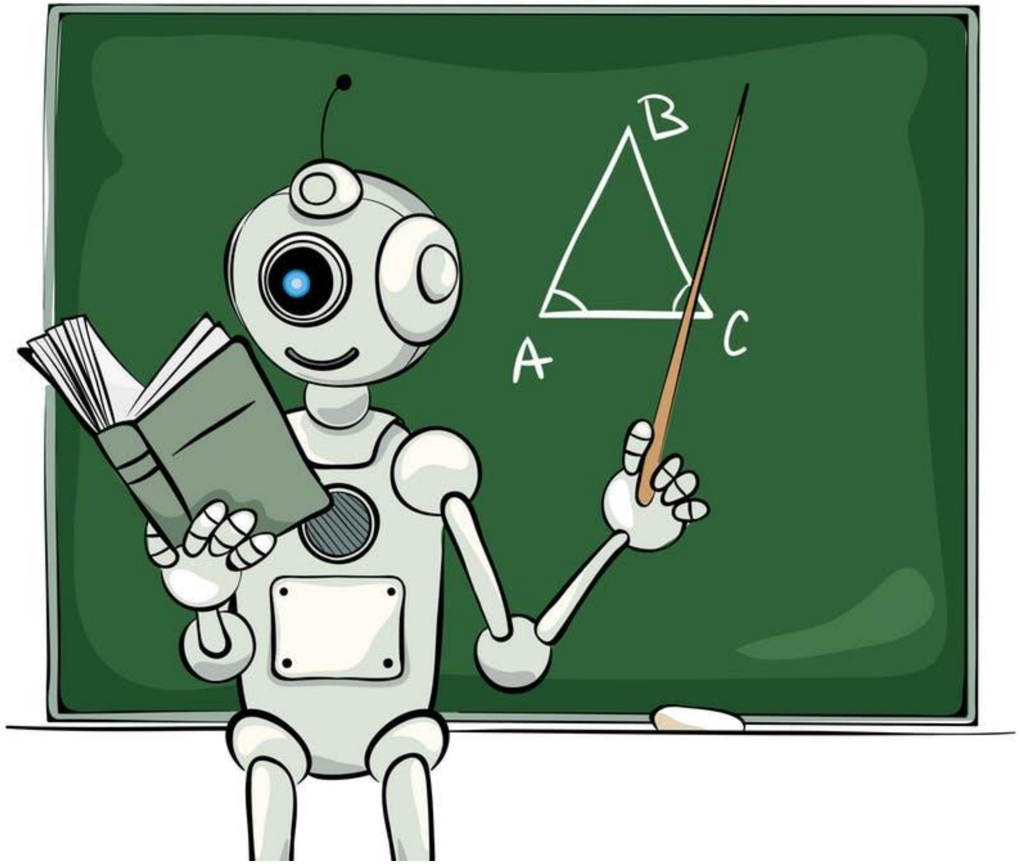
  "I liked to parade my learning in front of my sisters"

  synonyms: scholarship, knowledge, education, erudition, culture, intellect, academic attainment, acquirements, enlightenment, illumination, edification, book learning, insight, information, understanding, sageness, wisdom, sophistication; More

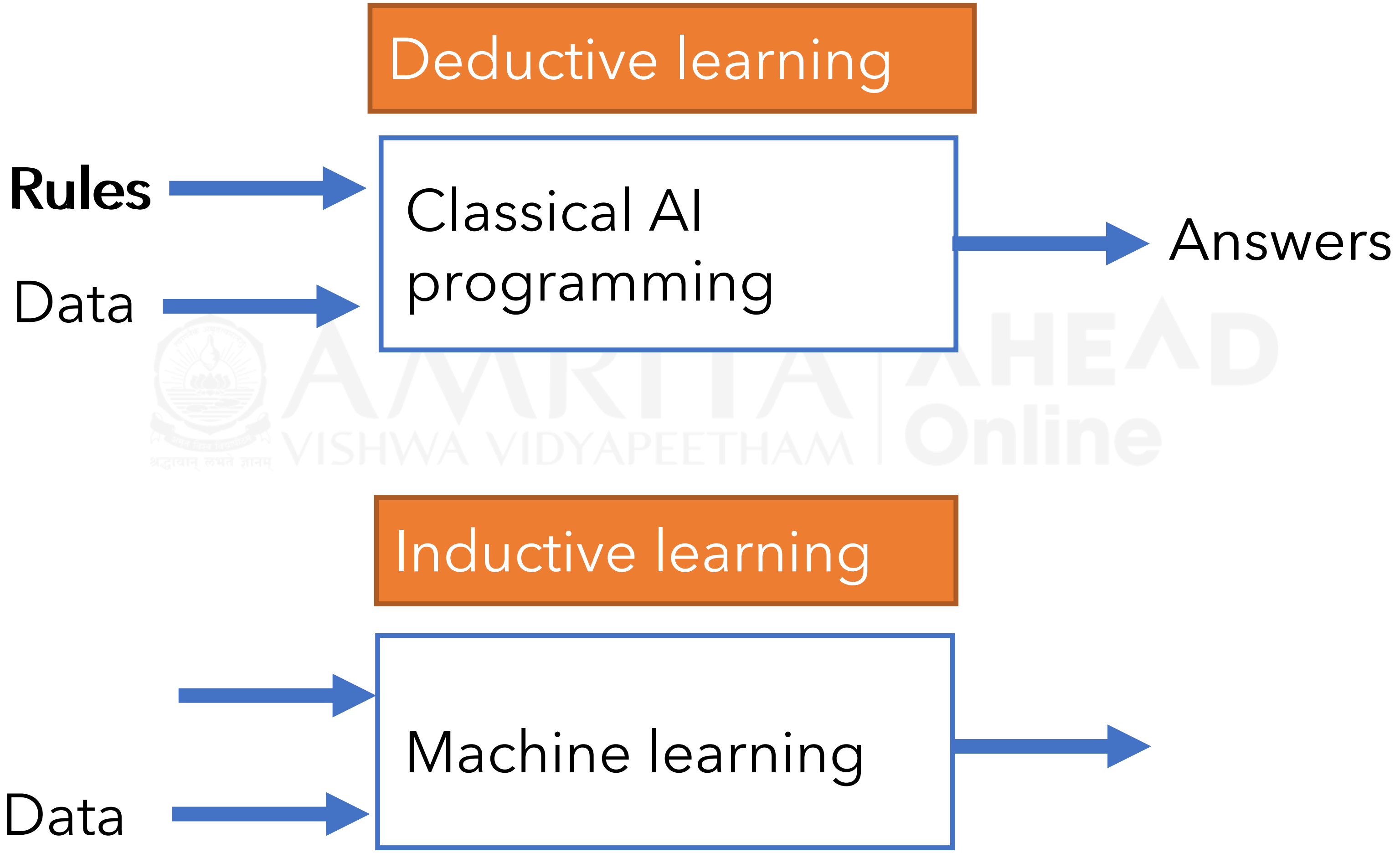
  antonyms: ignorance

How a Computer Learns

- A computer **experiences** the world with **data**
- A computer **studies** by applying **algorithms** to its data
- We **teach** computers by programming explicit concepts we are aware of
  - Domain knowledge

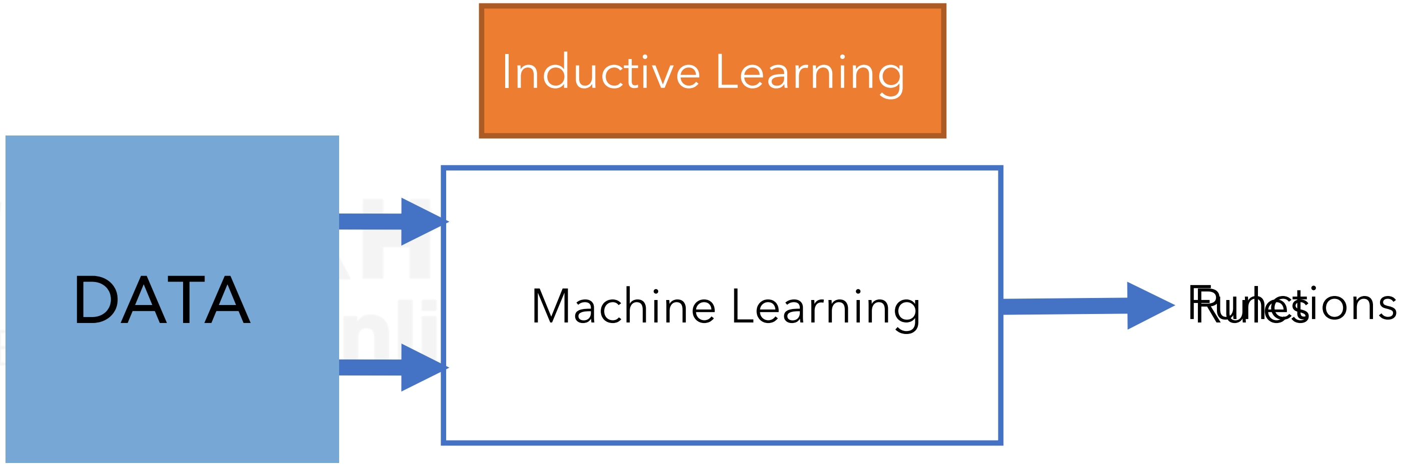


Deductive vs Inductive Learning



Machine Learning is Inductive Learning

- “Answers” are called labels
- “Rules” are really functions
- The functions are learned by applying algorithms to data, we call this **machine learning**
- Attempting to approximate a universal “true f(x)”
- Limited by the inputs: **data**



Conclusion

- Machine learning is inductive learning
  - It learns functions from data.
- Machine learning works by applying algorithms to data.
- All machine learning problems are limited by the data they have access to.
  - "Garbage in, Garbage out"

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Numpy

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Matplotlib

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Pandas

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Scikit-Learn

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NumPy

Jupyter

matplotlib

Version 3.1.1

Scikit-Learn

pandas

$$y_{it} = \beta' x_{it} + \mu_i + \epsilon_{it}$$

Ana(Conda)

Juypter + Scikit

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Introduction

- Jupyter
- Scikit-learn

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jupyter

Jupyter

- Jupyter
- Ju – Pyt – ER
- Julia, Python, R
- Interactive “notebooks” that run a kernel

scikit learn

Scikit Learn

- The de-facto, standard library for “classical machine learning”
- Excellent website + documentation
- Helps you learn how to do machine learning
- Has *many* algorithms, but not all
- Excellent unified API

Conclusion

- Looked at two of the most important tools of trade

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Bagging

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Introduction

- Working in groups is better
- Bagging
- Bootstrapping
  - With replacement
  - Without replacement

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Working in a group is better

- Manage high variance by having an **ensemble** of classifiers that will vote together
  - If everyone votes we won't go radically into any direction
  - Everyone voting together *smooths* our decision function

Ensemble Strategy

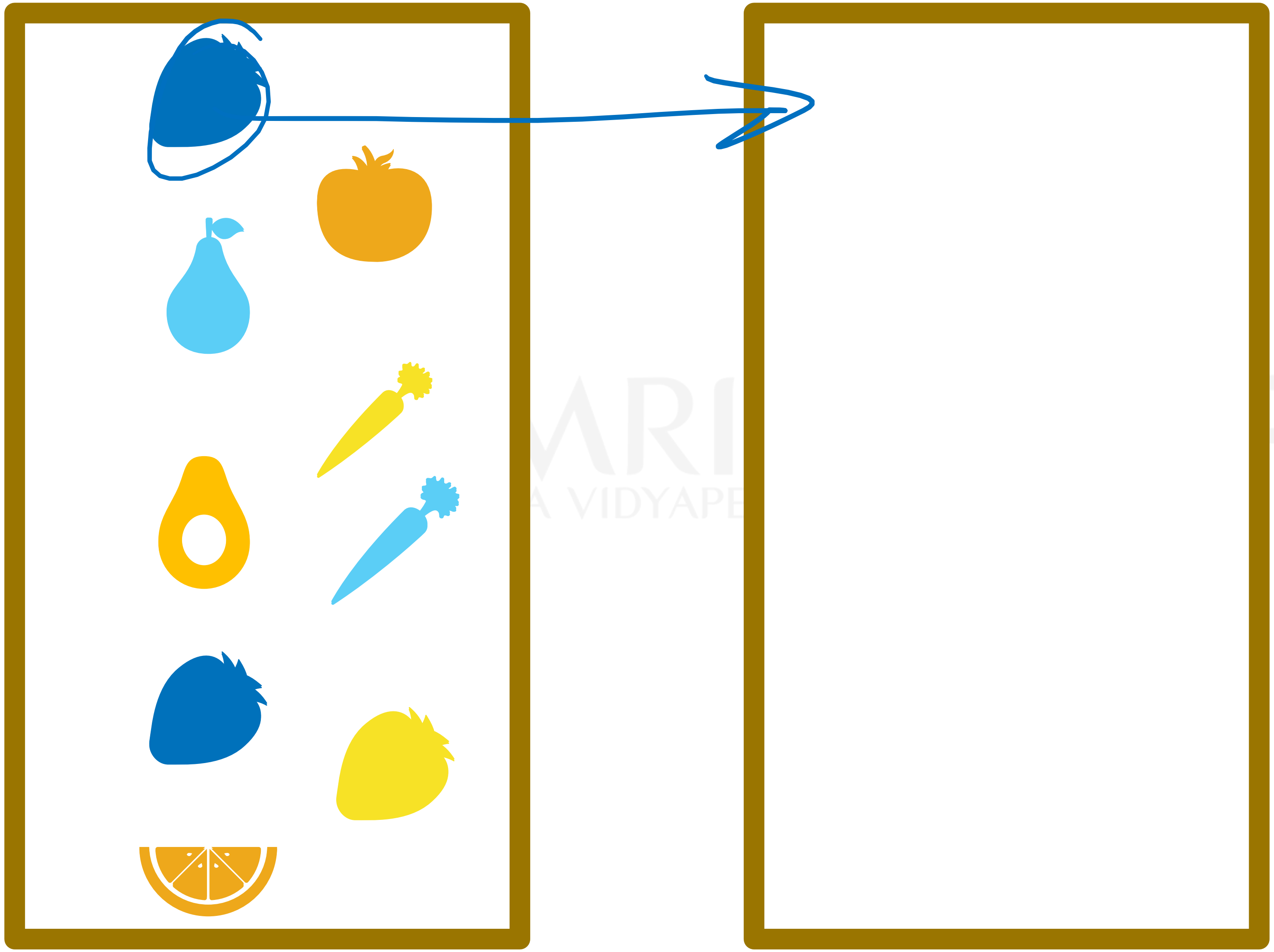
- What characteristics do we want the individual learners to have? The entire ensemble?
  - Each individual should have a different opinion
  - The ensemble should produce a good answer
- How do we make different opinions?

Bagging – Bootstrap Sampling

- Randomly “bootstrap” resample things to draw new distributions (make bags!)
- Original set: [apple_1, orange, melon, watermelon, apple_2, pineapple]
    - Make subsets:
    - Bag1: [apple_2, orange, melon, apple_1]
    - Bag2: [orange, melon, watermelon, apple_2]

Without Replacement

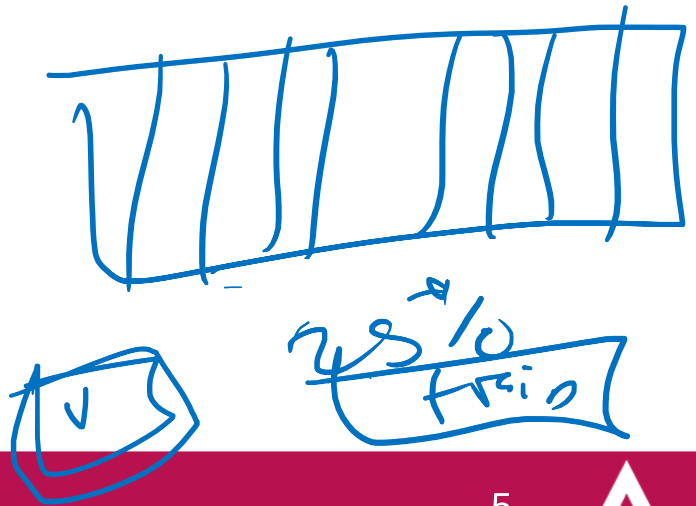
D1



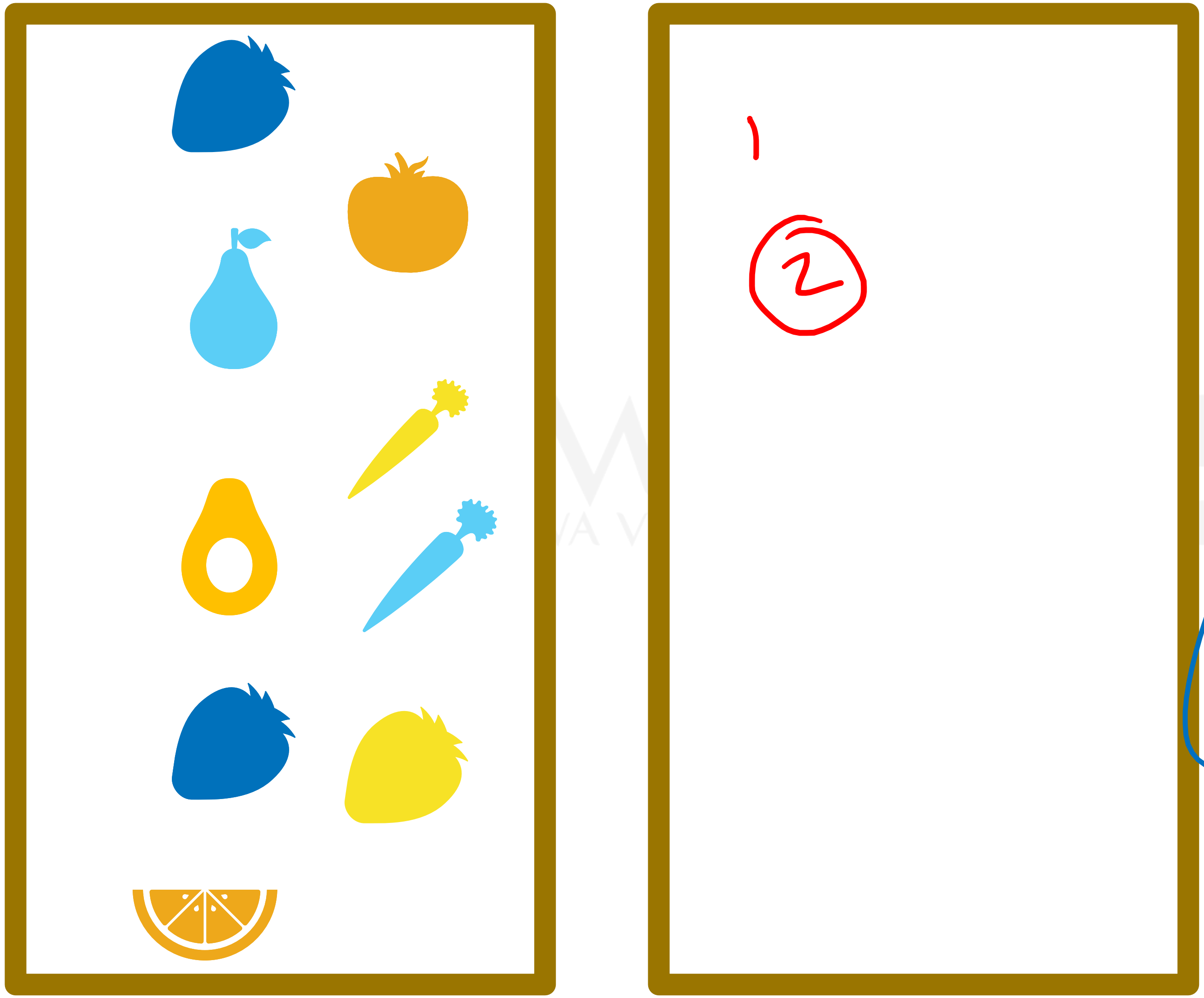
D1

- We finish with a new dataset that is a subset of the original dataset

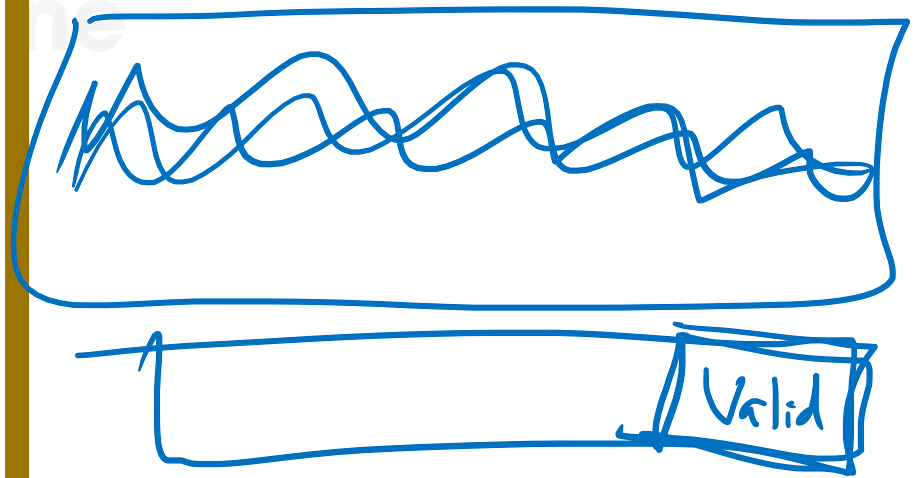
D2



With Replacement



- We finish with a new dataset that is the same size, but a variation of the original data



Conclusion

- Ensembles should have many opinions
- Bagging helps diversify opinions
- Bootstrap sampling
  - w/ replacement
  - w/o replacement

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